

# Filippo Maggioli, Ph.D.

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## Short Bio

### 📌 Research profile

I am a Tenured Assistant Professor at *Pegaso University*. Previously, I was Postdoctoral Researcher at *University of Milano-Bicocca* in the *DIG AIR* research lab led by Simone Melzi, a Postdoctoral Researcher at *Sapienza – University of Rome* in the *GLADIA* research lab led by Emanuele Rodolà, and a Research Intern at the *King Abdullah University of Science and Technology (KAUST)* in the *VCC* research lab led by Peter Wonka. I received my Ph.D. in Computer Science at *Sapienza – University of Rome* (2023), where I also graduated in Computer Science (2019).

I work on geometry processing, spectral geometry, and 3D shape analysis, but I am an active researcher also in other fields of computer graphics, such as procedural shading and physical simulation.

I regularly serve in the program committee of international conferences as chair and reviewer, and I maintain worldwide collaborations with researchers from other institutions and countries.

### 📌 Research interests

Geometry Processing; Spectral Geometry; 3D Shape Analysis; Procedural Texturing; Simulation of Natural Phenomena; Numerical Linear Algebra.

### 📌 Author profiles

ORCID:  0000-0001-8008-8468

Google Scholar ID: VN1fbwUAAAAJ *h*-index: 6 *i10*-index: 5

Scopus Author ID: 57216313662 *h*-index: 5 *i10*-index: 1

## Academic Appointments

|                     |                                                                                                                                                                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mar 2025 – Present  | 📌 <b>Assistant professor - tenure track.</b> Pegaso University<br>Department of Computer Science and Information Technologies.                                                                                                                     |
| Dec 2023 – Feb 2025 | 📌 <b>Postdoctoral researcher.</b> University of Milano-Bicocca<br>Member of the <i>DIG AIR</i> research lab.<br>Advisor: prof. Simone Melzi.<br>Research activity on computational and spectral geometry.                                          |
| Apr 2024 – Feb 2025 | 📌 <b>Adjunct professor.</b> Pegaso University<br>Undergrad courses on <i>Computer Architecture</i> and <i>Networking and Cybersecurity</i> .<br>Supervision of BSc students during the development of their theses.                                |
| Aug 2023 – Nov 2023 | 📌 <b>Postdoctoral researcher.</b> Sapienza – University of Rome<br>Member of the <i>Gladia</i> research lab.<br>Advisor: prof. Emanuele Rodolà.<br>Research activity on computational geometry, spectral geometry, and numerical linear algebra.   |
| Sep 2022 – Jan 2023 | 📌 <b>Research internship.</b> King Abdullah University of Science and Technology<br>Member of the <i>VCC</i> research lab.<br>Advisor: prof. Dominik L. Michels.<br>Research activity on simulation of natural phenomena in agricultural settings. |
| Mar 2021 – Jul 2021 | 📌 <b>Teaching assistant.</b> Sapienza – University of Rome<br>Undergrad course on <i>Introduction to Algorithms</i> .                                                                                                                              |

## Education

- 2019 – 2023     **Ph.D. in Computer Science.** Sapienza – University of Rome.  
Advisor: prof. Emanuele Rodolà.  
Thesis title: *Scalable geometry processing for computer graphics applications.*  
**Honourable mention at EG-Italy Award for PhD Thesis in Computer Graphics.**
- 2018 – 2019     **M.Sc. in Computer Science.** Sapienza – University of Rome.  
Advisor: prof. Emanuele Rodolà.  
Thesis title: *Time-efficient function reconstruction via Laplacian eigenproducts.*
- 2014 – 2017     **B.Sc. in Computer Science.** Sapienza – University of Rome.  
Advisor: prof. Enrico Tronci.  
Thesis title: *Modeling of biological pathways with systems of differential-algebraic equations.*

## Courses and Schools

- Jul 2022     **IRDTA DeepLearn 2022 Summer.**  
6th International Gran Canaria School on Deep Learning. *Las Palmas de Gran Canaria, Spain.*
- Jul 2021     **ACDL 2021.**  
Advanced Online & Onsite Course on Data Science & Machine Learning. *Pontignano, Italy.*

## Academic Service

- 2025     **Poster chair.** STAG  
Smart Tools and Applications in Graphics. *Genova, Italy*
-  **Member of program committee.** AICCSA  
ACS/IEEE International Conference on Computer Systems and Applications. *Doha, Qatar*
-  **Member of program committee.** ECAI  
European Conference on Artificial Intelligence. *Bologna, Italy*
- 2023     **Member of program committee.** TAG-ML  
ICML's workshop on Topology, Algebra, and Geometry in Machine Learning. *Honolulu, Hawaii*
- 2022     **Member of RCDC Conference Coffee Committee.** ACM SIGGRAPH RCDC  
ACM SIGGRAPH Research Career Development Committee. *Vancouver, Canada*
- 2021     **Event chair.** STAG  
Smart Tools and Applications in Graphics. *Rome, Italy*

## Teaching

- 2025 – Present     **Networking and Cybersecurity**  
Tenured assistant professor. Undergrad course. *Pegaso University*
-  **Algorithms and Data Structures**  
Tenured assistant professor. Undergrad course. *Pegaso University*
- 2025     **Functional Maps: a spectral approach to the alignment of embeddings**  
Ph.D. course. *Sapienza – University of Rome*
-  **Digital Skills for Teaching**  
Teaching qualification training course. *Pegaso University*
-  **Teaching Networking**  
Teaching qualification training course. *Pegaso University*
- 2024     **Networking and Cybersecurity**  
Adjunct professor. Undergrad course. *Pegaso University*

## Teaching (continued)

- 2021
  - **Computer Architecture**  
Adjunct professor. Undergrad course. *Pegaso University*
  - **Introduction to Algorithms**  
Teaching assistant. Undergrad course. *Sapienza – University of Rome*

## Invited, Conference, and Seminars Talks

- Nov 2025
  - **UV Parametrization via Topological Disk Segmentation of Surfaces**  
Smart Tools and Applications in Graphics, 2025. *Genoa, Italy*
- Jun 2025
  - **Volumetric Functional Maps**  
Sapienza – University of Rome, hosted by *E. Rodolà. Rome, Italy.*
- Nov 2024
  - **Scalable geometry processing in computer graphics applications**  
Smart Tools and Applications in Graphics, 2024. *Verona, Italy*
  - **TACO: a benchmark for connectivity-invariance in shape correspondence**  
Smart Tools and Applications in Graphics, 2024. *Verona, Italy*
  - **Efficient Generation of Multimodal Fluid Simulation Data**  
Smart Tools and Applications in Graphics, 2024. *Verona, Italy*
- Dec 2023
  - **A physically-inspired approach to the simulation of plant wilting**  
ACM SIGGRAPH Asia, 2023. *Sydney, Australia.*
- Oct 2022
  - **MoMaS: mold manifold simulation for real-time procedural texturing**  
Pacific Graphics (PG), 2022. *Kyoto, Japan.*
- May 2022
  - **Strassen's algorithm in practice**  
Sapienza – University of Rome, hosted by *R. Marin. Rome, Italy.*
- Aug 2021
  - **Efficiently parallelizable Strassen-based multiplication of a matrix by its transpose**  
International Conference on Parallel Processing (ICPP), 2021. *Chicago, Illinois, USA.*
- May 2021
  - **Orthogonalized Fourier polynomials for signal approximation and transfer**  
EUROGRAPHICS (EG), 2021. *Vienna, Austria.*

## Honours & Awards

- 2025
  - **STAG Best Paper Award – Honourable Mention**  
*UV Parametrization via Topological Disk Segmentation of Surfaces*  
Smart Tools and Applications in Graphics.
- 2024
  - **Matteo Dellepiane Award for PhD Thesis in Computer Graphics – Honourable Mention**  
The Italian Chapter of EuroGraphics (EG-Italy).

## Research Grants and Funding

- 2022
  - **Sapienza Research Starting Grant: Avvio alla Ricerca – Tipo 2**  
Principal investigator for the project *Enhancing Procedural Computer Graphics in Multimedia Applications with Fast Geometry Processing Techniques.*
- 2021
  - **Sapienza Research Starting Grant: Avvio alla Ricerca – Tipo 1**  
Principal investigator for the project *Automation of Casting Mold Design for Industrial Fabrication of Digital Objects.*
- 2020
  - **Sapienza Research Starting Grant: Avvio alla Ricerca – Tipo 1**  
Principal investigator for the project *GPU Fluid Simulation on Non-Euclidean Domains and Application for Simulation of Erosion Phenomena.*

## Reviewing Service

- 2025
- **Pacific Graphics.** Pacific Conference on Computer Graphics and Applications.
  - **ECAI.** European Conference on Artificial Intelligence.
  - **GCA.** IEEE Computer Graphics and Applications.
  - **ICCV.** International Conference on Computer Vision.
  - **CVPR.** IEEE/CVF Conference on Computer Vision and Pattern Recognition.
  - **TPAMI.** Transactions on Pattern Analysis and Machine Intelligence.
  - **CAVW.** Computer Animation & Virtual Worlds.
  - **EUROGRAPHICS.** Annual Conference of the European Association for Computer Graphics.
- 2024
- **ToG.** ACM Transaction on Graphics.
  - **ACCV.** Asian Conference on Computer Vision.
  - **Pacific Graphics.** Pacific Conference on Computer Graphics and Applications.
  - **ECCV.** European Conference on Computer Vision.
  - **EUROGRAPHICS.** Annual Conference of the European Association for Computer Graphics.
  - **CGF.** Computer Graphics Forum.
  - **TVCG.** IEEE Transactions on Visualization and Computer Graphics.
- 2023
- **Pacific Graphics.** Pacific Conference on Computer Graphics and Applications.
  - **TAG-ML.** ICML's workshop on Topology, Algebra, and Geometry in Machine Learning.
  - **ICCV.** International Conference on Computer Vision.
  - **NeurReps.** NeurIPS' workshop on Symmetry and Geometry in Neural Representations.
  - **ICIAP.** International Conference on Image Analysis and Processing.
- 2022
- **EUROGRAPHICS.** Annual Conference of the European Association for Computer Graphics.

## Supervision and Mentoring

- Ph.D.
- Giulio Viganó (University of Milano-Bicocca) – Internal supervisor (not formal advisor)
  - Francesca Maccarone (University of Milano-Bicocca) – Internal supervisor (not formal advisor)
  - Daniele Baieri (Sapienza – University of Rome) – Internal supervisor (not formal advisor)
  - Francesco De Canio (Sapienza – University of Rome) – Internal supervisor (not formal advisor)
- M.Sc.
- Giorgio Longari (University of Milano-Bicocca) – Internal supervisor (not formal advisor)
  - Roberta Giorgi (Sapienza – University of Rome) – Internal supervisor (not formal advisor)
  - Thesis advisor of 5 students (Pegaso University)
- B.Sc.
- Simone Pedico (University of Milano-Bicocca) – Thesis co-advisor
  - Alessio Tosato (University of Milano-Bicocca) – Thesis co-advisor
  - Alireza Alipanah (Sharif University of Technology) – Internal supervisor (not formal advisor)
  - Daniele Solombrino (Sapienza – University of Rome) – Internal supervisor (not formal advisor)
  - Thesis advisor of 11 students (Pegaso University)

## Skills

- Languages
- Italian (mother tongue), English (professional proficiency).
- Interpersonal
- Adaptability to work independently and with(in) a team. Capability of supervising and communicating efficaciously. Excellent organizational and teaching abilities.
- Programming
- Proficient in C/C++ and MATLAB. Advanced knowledge of GPU programming with CUDA, GLSL, and HLSL. Knowledge of Python and C#.

## Skills (continued)

- Tools  Expert with the mesh processing software *MeshLab* and the rendering engine *Blender*. Advanced knowledge of the game engines *Unreal Engine 4 and 5* and *Unity 3D*. Familiar with software for raster (*GIMP*) and vector (*InkScape*) 2D graphics.

## Research Publications

### Journal Articles

- 1 D. Marin, **F. Maggioli**, S. Melzi, S. Ohrhallinger, and M. Wimmer, “Reconstructing curves from sparse samples on riemannian manifolds,” *Computer Graphics Forum*, vol. 43, no. 5, e15136, 2024.
- 2 **F. Maggioli**, R. Marin, S. Melzi, and E. Rodolà, “Momas: Mold manifold simulation for real-time procedural texturing,” *Computer Graphics Forum*, vol. 41, no. 7, pp. 519–527, 2022.
- 3 L. Moschella, S. Melzi, L. Cosmo, **F. Maggioli**, O. Litany, M. Ovsjanikov, L. Guibas, and E. Rodolà, “Learning spectral unions of partial deformable 3d shapes,” *Computer Graphics Forum*, vol. 41, no. 2, pp. 407–417, 2022.
- 4 **F. Maggioli**, S. Melzi, M. Ovsjanikov, M. M. Bronstein, and E. Rodolà, “Orthogonalized fourier polynomials for signal approximation and transfer,” *Computer Graphics Forum*, vol. 40, no. 2, pp. 435–447, 2021.
- 5 **F. Maggioli**, T. Mancini, and E. Tronci, “Sbml2modelica: Integrating biochemical models within open-standard simulation ecosystems,” *Bioinformatics*, vol. 36, no. 7, pp. 2165–2172, 2020.

### Conference Proceedings

- 1 D. Baieri, **F. Maggioli**, E. Rodolà, S. Melzi, and Z. Löhner, “Implicit-arap: Efficient handle-guided neural field deformation via local patch meshing,” in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- 2 A. Ferraris, F. Leveni, D. Baieri, **F. Maggioli**, S. Melzi, and L. Magri, “Geometric aware local optimization for robust primitive fitting,” in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2025.
- 3 **F. Maggioli**, D. Baieri, E. Rodolà, and S. Melzi, “Rematching: Low-resolution representations for scalable shape correspondence,” in *Computer Vision – ECCV 2024*, Cham: Springer Nature Switzerland, 2025, pp. 183–200, ISBN: 978-3-031-72913-3.
- 4 **F. Maggioli** and S. Melzi, “Uv parametrization via topological disk segmentation of surfaces,” in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2025.
- 5 D. Baieri, D. Crisostomi, S. Esposito, **F. Maggioli**, and E. Rodolà, “Efficient generation of multimodal fluid simulation data,” in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2024.
- 6 F. Maccarone, G. Longari, G. Viganò, D. Peruzzo, **F. Maggioli**, and S. Melzi, “S4a: Scalable spectral statistical shape analysis,” in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2024.
- 7 S. Pedico, S. Melzi, and **F. Maggioli**, “Taco: A benchmark for connectivity-invariance in shape correspondence,” in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2024.
- 8 **F. Maggioli**, J. Klein, T. Hädrich, E. Rodolà, W. Palubicki, S. Pirk, and D. L. Michels, “A physically-inspired approach to the simulation of plant wilting,” in *SIGGRAPH Asia 2023 Conference Papers*, 2023, pp. 1–8.

- 9 **F. Maggioni**, D. Baieri, S. Melzi, and E. Rodolà, “Newton’s fractals on surfaces via bicomplex algebra,” in *ACM SIGGRAPH 2022 Posters*, 2022, pp. 1–2.
- 10 V. Arrigoni, **F. Maggioni**, A. Massini, and E. Rodolà, “Efficiently parallelizable strassen-based multiplication of a matrix by its transpose,” in *Proceedings of the 50th International Conference on Parallel Processing*, 2021, pp. 1–12.

## Pre-prints

- 1 **F. Maggioni**, M. Livesu, and S. Melzi, “Volumetric functional maps,” arXiv preprint arXiv:2506.13212, 2025.
- 2 L. Olearo, G. Viganò, D. Baieri, **F. Maggioni**, and S. Melzi, “Fuse: A flow-based mapping between shapes,” arXiv preprint arXiv:2511.13431, 2025.
- 3 **F. Maggioni**, D. Baieri, Z. Lähner, and S. Melzi, “Sshade: A framework for scalable shape deformation via local representations,” arXiv preprint arXiv:2409.17961, 2024.
- 4 D. Baieri, S. Esposito, **F. Maggioni**, and E. Rodolà, “Fluid dynamics network: Topology-agnostic 4d reconstruction via fluid dynamics priors,” arXiv preprint arXiv:2303.09871, 2023.